

Here Comes the Homology

Lecture 5 - CMSE 890

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Goals

Goals for today:

- Ch 2.4: Cycles and Boundaries

Section 1

Linear Algebra Review & Homology

Definition

A field $(k, +, \cdot)$ is a set k with two operations $+$ and $\cdot$ such that for any $a, b, c \in k$:

- Closure: $a+b \in k$ $a \cdot b \in k$
- Commutativity:
 - ▶ $a+b = b+a$
 - ▶ $a \cdot b = b \cdot a$
- Associativity:
 - ▶ $(a+b)+c = a+(b+c)$
 - ▶ $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- Identity:
 - ▶ $0_k \Rightarrow a + 0_k = a$
 - ▶ $1_k \Rightarrow a \cdot 1_k = a$

- Inverses:

- ▶ $a \quad \exists -a \text{ st. } a+(-a) = 0_k$
- ▶ $a \quad \exists \frac{1}{a} \text{ st. } a \cdot (\frac{1}{a}) = 1_k$

- Distributivity:

$$a(b+c) = ab+ac$$

Examples:

$$(\mathbb{R}, +, \cdot)$$

Non-ex: $(\mathbb{Z}, +, \cdot)$

$$(\mathbb{Z}_2, +, \cdot) = \{0, 1\}$$

$$\mathbb{Z}_p = \{0, 1, 2, \dots, p-1\}$$

$$p=7 \quad 5+6 \bmod 7 = 4$$

$$0+0=0$$

$$0+1=1$$

$$1+1=0$$

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 1 = 1$$

Vector space

Definition

A vector space over a field k is a set V with vector addition and scalar multiplication such that

- Associative +: $(v+w)+x = v+(w+x)$

- Commutative +: $v+w = w+v$

- Identity

- + $0_v \in V$ $w + 0_v = w$

- $1_k \in k$ $1_k \cdot v = v$

- Inverses:

- + Given $v \in V$, $\exists -v$ st $v + (-v) = 0_v$

- Scalar vs. field mult:

- Distributivity:

- ▶ $(a+b)v = av + bv$

- ▶ $a(v+w) = av + aw$

Examples:

$$V = \mathbb{R}^2$$

$$k = \mathbb{R}$$

$$(0+1)(v) = 0 \cdot v + 1v$$

$$1 \cdot v$$

$$v = 0 \cdot v + 1 \cdot v$$

Basis

Definition

A basis for a vector space V is a collection of vectors $\{b_\alpha\}_{\alpha \in A}$ such that

- they are linearly independent and,

$$\text{for any } t_\alpha \in k \quad \sum_{\alpha \in A} t_\alpha b_\alpha = 0 \Rightarrow t_\alpha = 0 \quad \forall \alpha$$

- they span V .

$$\text{Given any } v \in V, \quad \exists \{t_\alpha\}_\alpha \text{ s.t. } \sum t_\alpha b_\alpha = v$$

Goal

Build a vector space from a simplicial complex! (Next time)

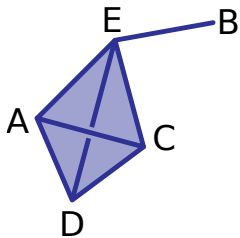
p -Chains

Let K be a simplicial complex and fix a dimension p .

A p -chain is a finite formal sum of p -simplices in K , written

$$\alpha = \sum_i a_i \sigma_i$$

σ_i is a p -simplex
elements of $k = \mathbb{Z}_2$



Example 2-chain

$$\alpha = AC + CE + BE + 0 \cdot AD + 0 \cdot CD + 0 \cdot DE + 0 \cdot AE$$

$$\beta = AD + AC + CD$$

Addition of chains

p -chains are added component-wise: if $\alpha = \sum a_i \sigma_i$ and $\beta = \sum b_i \sigma_i$, then

$$\alpha + \beta = \sum (a_i + b_i) \sigma_i$$

$$\begin{array}{l} \alpha = AC + CE + BE \\ \beta = AD + AC + CD \end{array} \left. \vphantom{\begin{array}{l} \alpha = AC + CE + BE \\ \beta = AD + AC + CD \end{array}} \right\} \begin{array}{l} \alpha + \beta = 1 \cdot AD + (1+1)AC + 1 \cdot CE + 1 \cdot BE + 1 \cdot CD \\ = AD + CE + BE + CD \end{array}$$

Chain group

The collection of p -chains with addition is called the p^{th} -chain group (vector space), $C_p(K)$.

Some checks

- Associative +: $(\alpha + \beta) + \gamma = \sum (a_i + b_i) \sigma_i + \sum c_i \sigma_i = \sum (a_i + b_i + c_i) \sigma_i = \alpha + (\beta + \gamma)$
- Commutative +:
- Zero element
- Inverses: $\sum 0 \cdot \sigma_i = 0$
Given $\alpha = \sum a_i \sigma_i$ $-\alpha = \sum (-a_i) \sigma_i$

$$\alpha = AC + CE + BE$$

$$\text{In } \mathbb{Z}_2! \quad -\alpha = \alpha$$

$$\text{In } \mathbb{R} \quad -\alpha = -AC - CE - BE$$

Linear Transformations

A linear transformation between two vector spaces V and W is a map $T : V \rightarrow W$ such that the following hold:

- $T(v+w) = T(v) + T(w)$
- $T(a \cdot v) = a \cdot T(v)$

$$\sum_{i=1}^n a_i b_i$$

Matrix representation: T picked a basis for V and W
 $\{b_1, \dots, b_n\} \sim \{c_i\}$

$$\begin{pmatrix} | & | & & | \\ T b_1 & T b_2 & \dots & T b_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \\ \\ \end{pmatrix}$$

Boundary maps¹

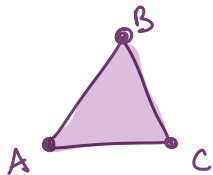
$$C_3(K) \longrightarrow C_2(K)$$

$$0 \longmapsto 0$$

$$\partial_p : C_p(K) \longrightarrow C_{p-1}(K)$$

$$\sigma = [v_0, \dots, v_p] \mapsto \sum_{j=0}^p [v_0, \dots, \hat{v}_j, \dots, v_p]$$

basis $\{AB, AC, BC\}$
remove $\hat{v}_j$ from the list



$$C_1(K) \longrightarrow C_0(K)$$

$$AB \longmapsto [\hat{A}, B] + [A, \hat{B}] = B + A$$

$$AC \longmapsto [\hat{A}, C] + [A, \hat{C}] = C + A$$

$$BC \longmapsto [\hat{B}, C] + [B, \hat{C}] = C + B$$

$$C_2(K) \longrightarrow C_1(K)$$

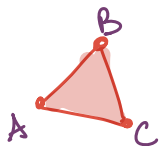
$$[A, B, C] \longmapsto [\hat{A}, B, C] + [A, \hat{B}, C] + [A, B, \hat{C}]$$

$$= BC + AC + AB$$

¹Warning: We are assuming $\mathbb{Z}_2$ coefficients from now on!

Checking linearity

$$\partial_p(\alpha + \beta) = \partial_p(\alpha) + \partial_p(\beta); \quad \alpha = \sum a_i \sigma_i, \quad \beta = \sum b_i \sigma_i$$



$$C_1(K) \xrightarrow{\partial_1} C_0(K)$$

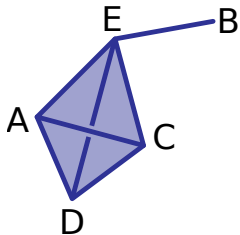
$$\alpha = AB + BC \longmapsto A + B + B + C = A + C$$

$$\beta = BC + AC \longmapsto B + C + A + C = A + B + C$$

$$\alpha + \beta = AB + 2BC + AC$$

$$= AB + AC \longmapsto A + B + A + C = B + C$$

Try it:



- $\partial_1([a, e]) =$

- $\partial_1([a, e] + [b, e]) =$

- $\partial_1([a, e] + [c, e] + [c, d] + [a, d]) =$

- $\partial_2([a, c, e] + [a, c, d]) =$

Ian's
homework